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Paleomagnetic data bearing on the origin of arcuate structures in the French Peak–Massachusetts Mountain area of southern Nevada

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ABSTRACT

The eastern margin of the Yucca Flat basin, in southern Nevada, is bounded by north-northwest–striking tilted fault blocks of the Halfpint Range whose strikes curve as much as 90° clockwise into east-northeast strikes in the French Peak–Massachusetts Mountain (FPMM) area. This pattern of arcuate structures has been attributed to clockwise drag along a postulated northwest-trending, right-lateral shear zone. The flexure model implies that rocks within the FPMM area were rotated strongly clockwise about a vertical axis. Directions of remanent magnetization of the middle Miocene Ammonia Tanks and Rainier Mesa Members of the Timber Mountain Tuff and of the Topopah Spring Member of the Paintbrush Tuff indicate no systematic vertical-axis rotation in the FPMM area and disprove the flexure model. After tilt correction, declinations of 29 site means obtained from the three ash-flow sheets in the FPMM area are not systematically different than declinations of 17 site means from the Halfpint Range or declinations of 16 site means from little-deformed mesa areas to the west. The paleomagnetic data thus indicate that structures in the FPMM area initiated with arcuate trends and were not originally straight elements that were rotated by right-lateral drag. The structures probably formed under the influence of spatially variable stress fields. The FPMM area lies in an accommodation zone between domains of oppositely tilted extensional fault blocks. Interaction between stress fields associated with propagating normal-fault zones may have been responsible for the arcuate structures in the FPMM area.

INTRODUCTION

The relative contributions of horizontal- and vertical-axis rotations to strain during continental extension commonly are only poorly known. Horizontal-axis rotations (tilts) typically are associated closely with normal faults, whereas vertical-axis rotations may be associated with strike-slip faults. The relative role of tilting and vertical-axis rotation is a pertinent question in the southern Great Basin area of the Basin and Range province of western North America, because here Cenozoic extension was accommodated by coeval normal faulting and strike-slip faulting (Wright, 1976; Wernicke and others, 1988). Tilting of normal-fault blocks accompanied fault offset and was of consistent direction over large areas (Stewart, 1980). Right-lateral strike-slip faults of northwest strike (for example, Las Vegas Valley shear zone, Longwell, 1974) and left-lateral strike-slip faults of northeast strike (for example, Lake Mead shear zone, Anderson, 1973) are also common in the region (Fig. 1). Rotations about vertical axes are known or inferred to be spatially associated with some of these strike-slip fault zones. Strong clockwise bending of rocks that has been documented in the southern Las Vegas Range adjacent to the Las Vegas Valley shear zone (Longwell, 1974; Nelson and Jones, 1987) is a prominent example. In the southern part of the Nevada Test Site and adjoining areas (Fig. 1), Burchfiel (1965), Ekren (1968), and Barnes and others (1982) attributed clockwise deflections of structures in Paleozoic rocks of the Specter Range and Spotted Range areas to strong clockwise vertical-axis rotation related to right-lateral movements along northwest-striking shear zones. Likewise, post–13 Ma clockwise vertical-axis rotation documented for southern Yucca Mountain (Rosenbaum and others, 1991) may also reflect drag along or above a right-lateral shear zone.

On the southeast margin of the Yucca Flat basin, Miocene volcanic rocks and faults that offset these rocks change strike from north-northwest in the Halfpint Range to east-northeast south of the French Peak area of Massachusetts Mountain, and to northeast in the southern part of Massachusetts Mountain (Figs. 2 and 3). Carr (1984) attributed the pattern of arcuate structures in the French Peak–Massachusetts Mountain (FPMM) area to clockwise flexure of originally north-northwest–trending structures above a northwest-trending, right-lateral shear zone (Fig. 4). The postulated shear zone extended about 40 km from the Spotted Range northwestward across northern Frenchman Flat and southern Yucca Flat. It was presumed to be one of a broad system of right-lateral faults (Carr, 1984) that constitute the Walker Lane belt (Carr, 1984; Stewart, 1988). The postulated structure was termed the "Yucca-Frenchman shear and flexure zone." The magnitude of apparent clockwise flexure above the shear zone would require several kilometers of right-lateral offset on the shear zone.

The FPMM area also lies within a generally east-west–striking accommodation zone (Bosworth and others, 1986; Faulds and others, 1988) between tilt domains of fault blocks (Fig. 2). Fault blocks north of the FPMM area (Halfpint Range, beneath Yucca Flat) were tilted west, whereas fault blocks to the south (southern Massachusetts Mountain, beneath Frenchman Flat) were tilted east or southeast.

The flexure model for the Yucca-Frenchman shear and flexure zone requires that rocks in the FPMM area rotated clockwise by as much as 90° about a vertical axis. Alternatively, if structures in the FPMM area nucleated with arcuate form under the influence of a spatially variable stress field, rocks in the area might have experienced little vertical-axis rotation. To choose between these two models, paleomagnetism of three Miocene ash-flow sheets was studied to assess the sense and magnitude of vertical-axis rotation in the FPMM area.

GEOLOGIC SETTING

Rocks in the Nevada Test site area (Fig. 1) are predominantly either of Paleozoic or Tertiary age. Byers and others (1989) gave a histori-

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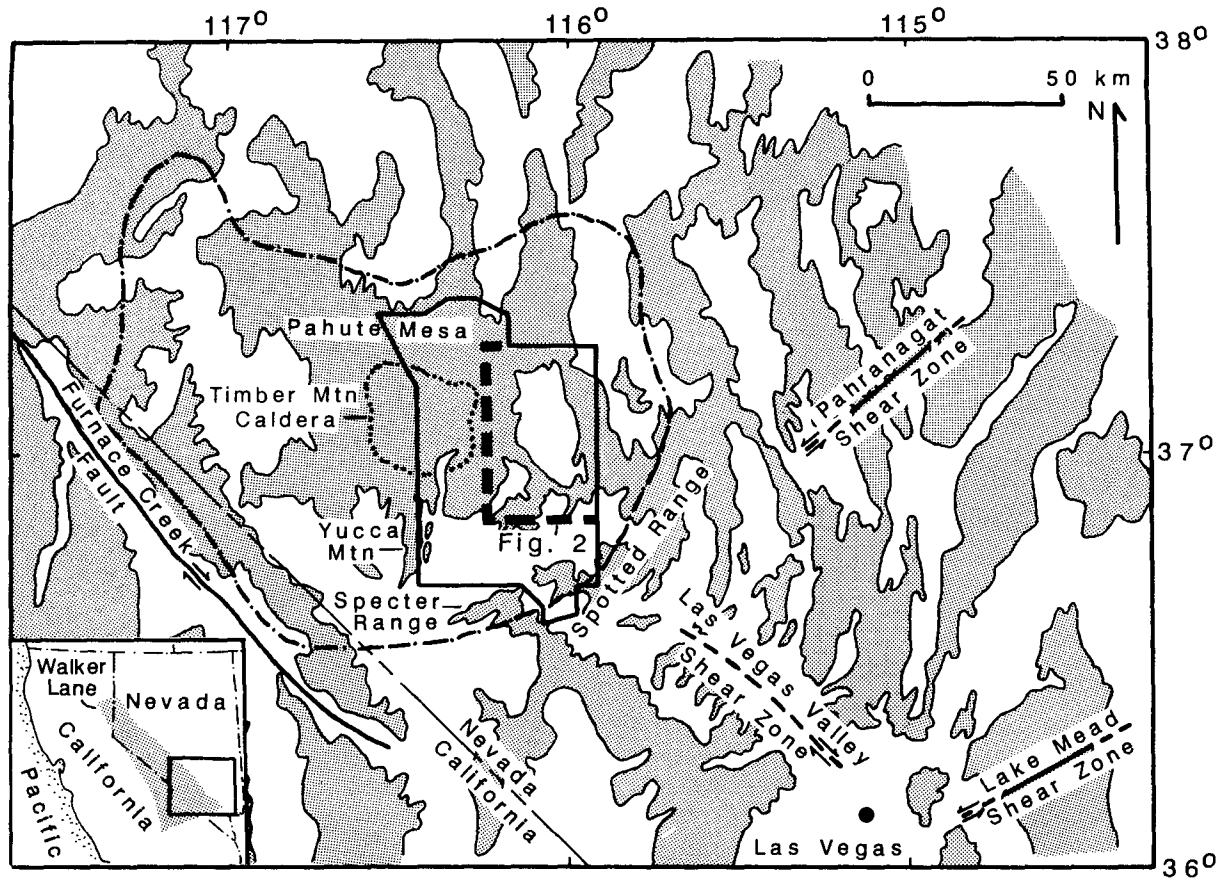


Figure 1. Regional tectonic map of part of the Basin and Range province, showing upland areas (shaded) and selected geographic and geologic features. Dash-dot line encloses the southwestern Nevada volcanic field. Solid line encloses the Nevada Test Site area.

cal review of geologic investigations in the area. Diverse sedimentary rocks of the Cordilleran miogeocline of western North America (Stewart and Poole, 1974; Poole, 1974) that range from late Precambrian to late Paleozoic in age are the oldest rocks in the area of Figure 2. Pre-Tertiary rocks are overlain by middle Miocene volcanic rocks of the southwestern Nevada volcanic field (Byers and others, 1976) that was active between about 16 Ma and 7.5 Ma (Sawyer and others, 1990). These volcanic rocks, which were derived from several eruptive centers in the central part of the volcanic field, are predominantly ash-flow tuffs with interlayered bedded tuffs and lava flows. Ammonia Tanks and Rainier Mesa Members of the Timber Mountain Tuff and the Topopah Spring Member of the Paintbrush Tuff are widely exposed in the FPMM area (Fig. 3). The Ammonia Tanks, Rainier Mesa, and Topopah Spring Members have yielded $^{40}Ar/^{39}Ar$ isotopic ages from alkali feldspars of 11.45 Ma, 11.6 Ma, and 12.8 Ma, respectively (Sawyer and others, 1990). Upper Miocene(?) to Quaternary alluvial deposits overlie middle Miocene or pre-Tertiary rocks in the structural basins.

Normal faults are the dominant middle Miocene tectonic structures of the area. In the Halfpint Range and Yucca Flat, the Miocene volcanic sequence and underlying pre-Tertiary rocks are broken by numerous normal faults having generally north to north-northwest strikes. These faults typically have down-to-the-east throws, and Miocene volcanic rocks within fault blocks are gently to moderately tilted (15°–30°) west. In contrast, south of French Peak, faults have predominantly down-to-the-west throws, and Miocene volcanic rocks within the fault blocks have east to southeast dips. Edges of the accommodation zone separating these two tilt-block domains (Fig. 2) are ill defined. Faults entering the zone generally lose throw gradually, and some change strike; rocks within fault blocks have dips that commonly shallow and change direction, defining broad flexures. Fault blocks entering the accommodation zone from either tilt domain may interfinger and extend past a median line of the zone.

Mesas composed of Miocene volcanic rocks (Pahute Mesa, Rainier Mesa, northern Yucca Mountain) surround the Timber Mountain caldera in the central part of the volcanic field west of Yucca Flat (Figs. 1 and 2). Although rocks in the mesas are broken by normal faults, these faults typically are more widely spaced and have less throw than faults in extended areas to the east (Halfpint Range, Yucca Flat). Strata in the mesas also dip more shallowly than in highly faulted areas to the east. Tectonic strain accommodated by faulting and tilting in the central part of the volcanic field is low in comparison to surrounding areas.

The onset of faulting that developed Yucca Flat and Frenchman Flat basins is not tightly constrained, but most deformation postdates deposition of members of the Timber Mountain Tuff. Fault blocks in the Halfpint Range and the FPMM area lack angular unconformities within the Miocene volcanic sequence. Moreover, drill-hole information indicates no structurally controlled thickening of members of the Timber Mountain Tuff beneath Yucca Flat as is found for younger alluvium (Carr, 1974). Basaltic dikes intrude normal faults in the central Halfpint Range (Byers and Barnes, 1967), indicating that faulting at this location began before ca. 8.5

Ma (Crowe and others, 1983). The Yucca fault in the center of Yucca Flat accommodated Holocene displacement (Carr, 1974). Major extension in the Yucca Flat and Frenchman Flat regions thus began sometime between ca. 11.4 Ma and 8.5 Ma and has continued, at least locally, into the Holocene.

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PALEOMAGNETIC TEST FOR VERTICAL-AXIS ROTATION

Test Procedures, Assumptions, Uncertainties

To test for vertical-axis rotation, tilt-corrected directions of remanent magnetization from the Ammonia Tanks and Rainier Mesa Members of the Timber Mountain Tuff and the Topopah Spring Member of the Paintbrush Tuff were compared for sites from the FPMM area, the Halfpint Range, and mesa regions west of Yucca Flat. The greatest number of sites were collected in the Rainier Mesa Member because of its wide distribution and good exposures.

Before declinations of paleomagnetic data are used to assess vertical-axis rotation, two facts should be known: (1) the predeformation orientation of the remanent magnetization and (2) the amount and direction of any tilt that affected the rocks. Comparison of remanent magnetization from different sites in an individual volcanic cooling unit yields estimates of relative tectonic rotation (Magill and others, 1982; Wells and Hillhouse, 1989; Rosenbaum and others, 1991). For this study, paleomagnetic data for sites from the mesa areas in the central part of the volcanic field were used as reference directions because these little-deformed areas were considered least likely to have been affected by local-scale vertical-axis rotation. Additionally, a time-averaged mean direction derived from numerous Miocene volcanic units indicates that the mesa areas have not been significantly rotated about a vertical axis with respect to the North American craton (Hudson and Rosenbaum, 1990).

The potential of pretectonic dispersion of remanent magnetizations within individual volcanic units must be considered when evaluating rotations. Detection of tectonic rotation is simplest if the direction of magnetization was consistent throughout a unit before deformation. A single direction of magnetization typically is recorded by thin, simple, ash-flow–tuff cooling units that cool rapidly with respect to paleosecular variation of the Earth's geomagnetic field (Dalrymple and others, 1965; Gromme and others, 1972; Reynolds, 1977). For thick, compound cooling units, however, care must be taken to determine whether rocks recorded different directions due to protracted cooling relative to paleosecular variation. Care also must be taken to assess whether later processes have altered the primary magnetization or added secondary magnetization in the tuffs (Reynolds, 1977; Geissman, 1980; Rosenbaum, 1986). Pertinent to this study, Rosenbaum (1986) documented a decrease in inclination of remanent magnetization in the Topopah Spring Member related to welding that occurred after acquisition of the magnetization in the interior of thick sections at Yucca Mountain.

Paleomagnetic data must be tilt corrected before their declinations can be compared to assess vertical-axis rotation. Tilt corrections for this study were performed by rotation of magnetization directions about the strike axis by the

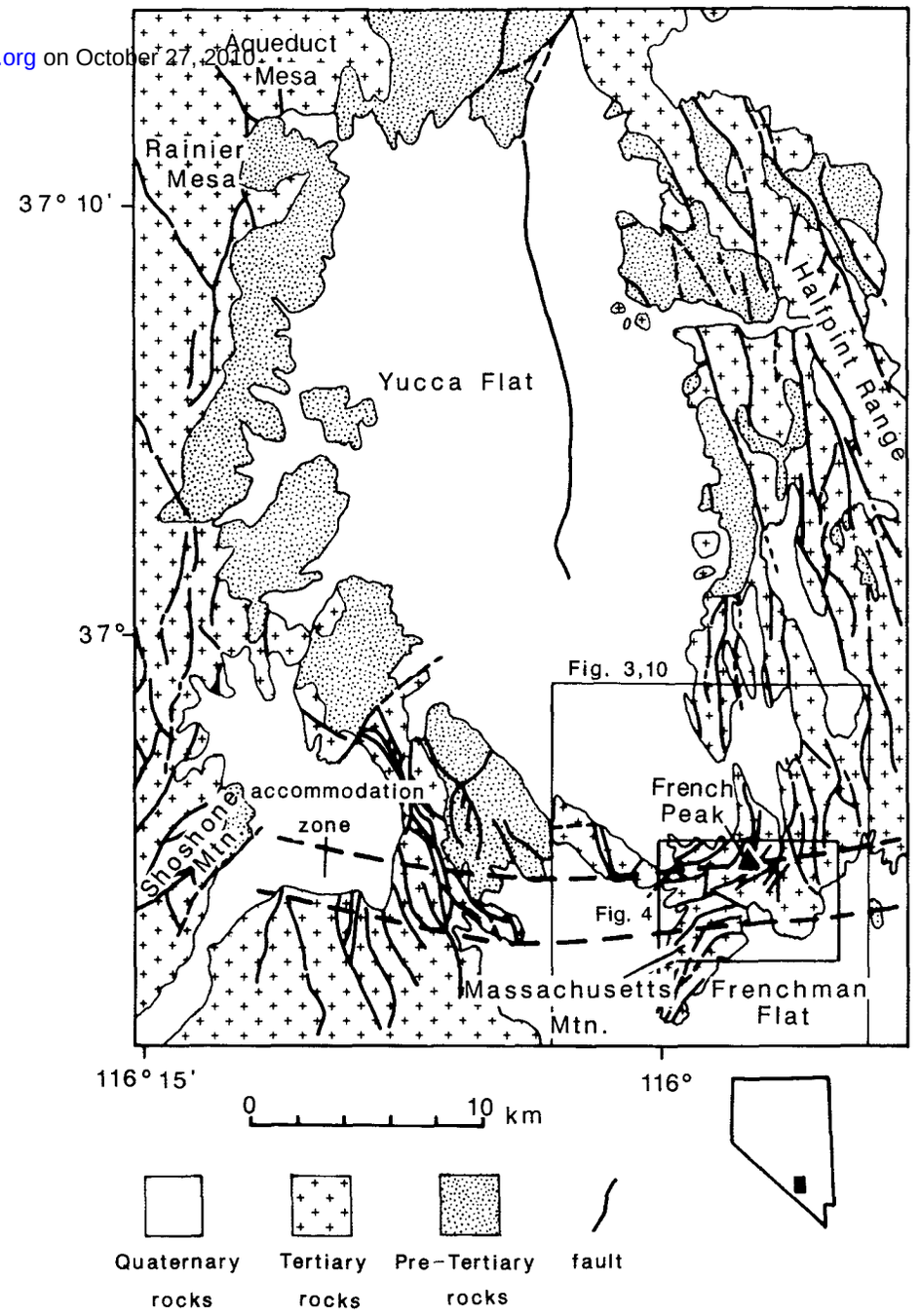


Figure 2. Generalized geologic map of the Yucca Flat area showing the location of the French Peak–Massachusetts Mountain area and an accommodation zone that separates fault blocks with west-dipping Tertiary strata to the north from fault blocks with east-dipping strata to the south.

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amount of dip for each site. A declination difference that remains after the tilt correction may, within uncertainty limits, be attributed to vertical-axis rotation. Without additional kinematic information, however, it can be difficult to assess whether such a declination difference reflects true vertical-axis rotation that occurred as an event separate from tilting or whether both the declination difference and dip for a site were caused by rotation about an inclined axis (MacDonald, 1980). Nevertheless, declination anomalies in tilt-corrected paleomagnetic data suggest that deformation involved more than tilt about a single-strike axis.

Paleomagnetic Data—Methods and Results

All samples for the study were collected with a portable rock drill and oriented with a clinometer and both magnetic and solar compasses. At most sites, seven to nine samples were collected from outcrops spread over several tens of square meters. At some sites, from 8 to 34 samples were collected in traverses across stratigraphic intervals of the Rainier Mesa or Ammonia Tanks Members to test for stratigraphically controlled variation of remanent magnetization. Samples from the Topopah Spring Member were collected from the upper few meters of the ash flow to minimize potential effects of subblocking-temperature welding and inclination flattening that, where previously detected (Rosenbaum, 1986), were most severe in the interior of the ash-flow sheet. Stratal attitudes of sites were estimated by measurements of compaction foliations, stratigraphic contacts, and boundaries between welding zones.

In the laboratory, samples were subjected to progressive alternating-field or thermal demagnetization. At least six samples from each site were subjected to alternating-field demagnetization (typically eight demagnetization steps per sample to peak inductions of 80 milliTesla [mT]). Representative samples (80) from 40 sites were thermally demagnetized. Demagnetization results were plotted on vector diagrams, and directions of components were determined by principal-component analysis (Kirschvink, 1980) for visually determined linear segments. Most site means and dispersion parameters (Table 1) were calculated using Fisher statistics (Fisher, 1953). For a few sites affected by lightning, site means were derived from intersecting remagnetization circles (Halls, 1976) using Bingham statistics (Onstott, 1980) and compared with mean directions derived using Fisher statistics on sample magnetizations for which the lightning-induced magnetization could be confidently removed. Mean directions derived using these two methods were typically very similar.

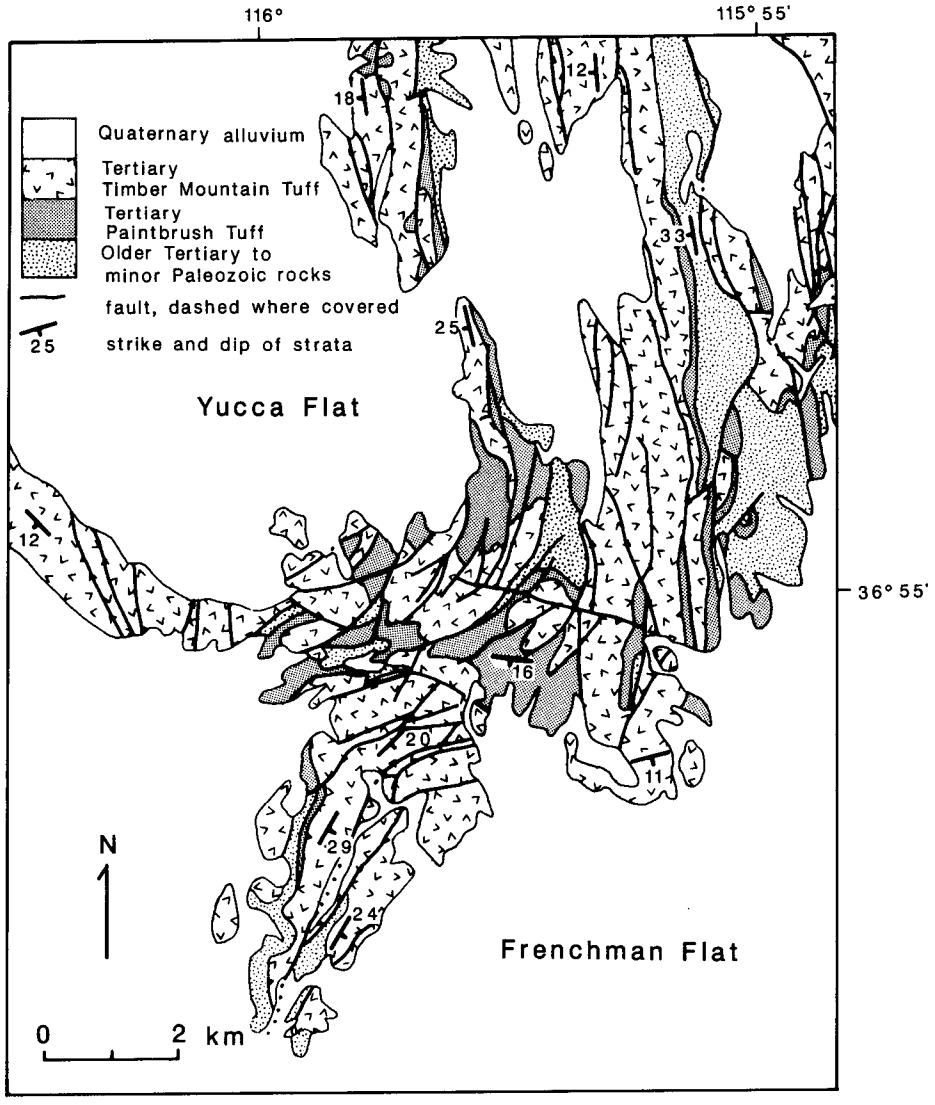


Figure 3. Geologic map of the French Peak–Massachusetts Mountain area generalized from Hinrichs and McKay (1965), Poole (1965), Poole and others (1965), and McKeown and others (1976).

The mean for site 9YF-37 (Table 1) is based on intersection of remagnetization circles.

Each ash-flow tuff carries a component of remanent magnetization interpreted to have been acquired during initial cooling after deposition. After removal of small-moment viscous magnetizations, further demagnetization isolated a single component of magnetization for most samples (Fig. 5). Typically, these components had well-grouped directions within a site. Values of the concentration parameter, k (Fisher, 1953), for site means typically exceeded 100 (Table 1).

Rosenbaum (1986) documented shallowing of inclinations of remanent magnetization from 70° to as low as 20° in the interiors of thick sections (>300 m) of Topopah Spring Member from drill holes on Yucca Mountain. This shallowing and an attendant increase in local-scale dispersion of magnetization was attributed to subblocking-temperature rotation of the magnetic carriers during welding. Thin stratigraphic sections and the upper 100 m of the thick sections of the Topopah Spring Member, however, apparently escaped significant subblocking-temperature distortions (Rosenbaum, 1986; Rosenbaum and others, 1991). Tilt-corrected means for sites from the Topopah Spring Member in this study have inclinations ranging from 55° to 76° and high k values (Table 1; Fig. 6), suggesting that they were not severely affected by subblocking-temperature rotations,

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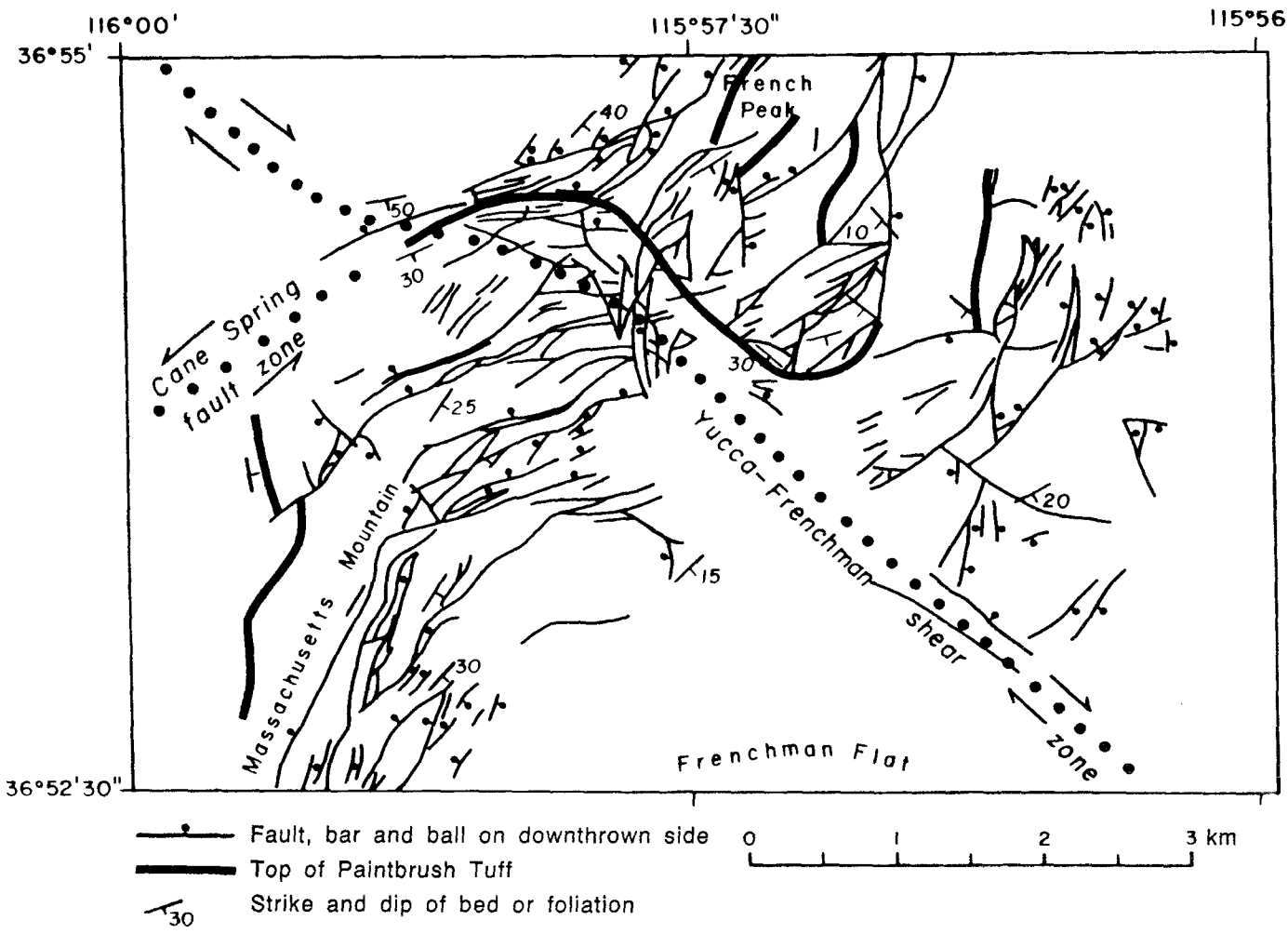


Figure 4. Yucca-Frenchman flexure and shear zone model from Carr (1984).

although some remanence flattening might be invoked to explain the inclinations as low as 55°.

Demagnetization and hysteresis properties indicate that fine-grained magnetite and perhaps maghemite carry most of the remanence in these ash-flow tuffs (Figs. 5A, 5B, 5C, 5D, and 7A). Laboratory unblocking temperatures in some samples that otherwise have magnetite-like properties extend above the 580 °C Curie temperature for magnetite to as high as 620 °C (Fig. 5D). This behavior, which has also been reported in other ash-flow tuffs (Gose, 1970; Gromme and others, 1972; Rosenbaum, 1986; Wells and Hillhouse, 1989; McIntosh, 1991), may reflect contributions from high-temperature stabilized maghemite (McIntosh, 1991) that grew as microcrystals from glass (Schlinger and others, 1988) during cooling (J. G. Rosenbaum, U.S. Geol. Survey, 1990, oral commun.).

Hematite is the dominant carrier of magnetization in a vapor-phase zone (Smith, 1960) in the upper Rainier Mesa Member in the FPMM and southern Halfpint Range areas. Properties indicative of hematite are resistance of remanence to alternating-field demagnetization (Fig. 5F), nonsaturation of isothermal remanent magnetizations at 930 mT and high minimum coercivities of remanence (Fig. 7B), and diminished magnetic susceptibilities and remanence moments. Unblocking of magnetization in such samples by 650 °C (Fig. 5E), below the 680 °C Curie point for hematite, suggests that the grain size of the hematite is small and that it may have been inherited by oxidation of precursor cubic-Fe-oxide microcrystals. Hematite probably formed during initial cooling of the Rainier Mesa sheet rather than later because of (1) its restriction to the capping vapor-phase zone, (2) low dispersion of hematite directions (for example, site 7YF-17, Table 1), indicating that oxidation was short-lived with respect to paleosecular variation, and (3) parallelism of hematite and magnetite components in samples in which oxidation was not complete (Fig. 5E).

After tilt correction, site means from each of the three units are less dispersed (Table 2), yielding a positive "fold" test (McElhinny, 1964) at the 99% confidence level for each unit. Dispersion of inclinations (McFadden and Reid, 1982), which is more appropriate to examine if the rocks were affected by vertical-axis rotation, also decreased significantly for each unit after tilt correction (at 99% confidence level based on F-test of k ratio; Watson, 1956). Positive fold tests and rock magnetic properties for the tuffs indicate that characteristic magnetizations in the three ash-flow tuffs predate deformation, probably represent thermoremanent or chemical-remanent magnetizations formed during initial cooling of each sheet and were little altered after initial cooling.

Systematic changes of remanent directions

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TABLE 1. PALEOMAGNETIC DATA FROM MIOCENE ASH-FLOW TUFFS IN SOUTHERN NEVADA

No.	Id.	Lat.	Long.	N/N_0	D_0	I_0	α_{95}	k	St/Dp	D_1	I_1
				Mesa areas							
Ammonia Tanks Member											
1	8YF-29	37.223	116.167	5/9	342.2	55.9	5.0	239	358/5	349.6	57.0
7	7YF-3	37.174	116.208	9/9	2.6	62.6	3.2	256	180/4	354.9	62.5
8	9YF-37	36.995	116.226	8/8*	354.0	63.0	2.5	· ·	205/2	350.8	61.9
	JR89-2	37.197	116.479	7/7	355.7	64.5	1.8	1118	220/2	352.9	63.1
	reference mean			4/4			3.4	722		351.9	61.1
Rainier Mesa Member											
2	7YF-16	37.128	116.171	29/34	162.3	-48.6	1.9	196	5/12	176.3	-51.9
3	7YF-5	37.203	116.203	7/7	169.9	-51.3	1.2	2475	180/2	170.5	-51.5
4	7YF-4	37.197	116.210	7/7	173.6	-53.4	2.4	650	180/4	168.4	-52.8
5	7YF-2	37.182	116.221	6/8	181.5	-49.0	3.0	487	163/6	170.1	-48.3
6	8YF-54	37.171	116.219	29/29	178.1	-47.6	2.7	101	142/9	168.9	-52.3
				18/18†	185.5	-45.2	2.0	293	142/9	173.0	-54.1
9	9YF-36	36.995	116.225	7/7	172.6	-52.8	2.2	745	205/2	170.4	-51.7
	reference mean			6/6			2.1	981		171.4	-51.7
Topopah Spring Member											
10	9YF-34	36.993	116.223	8/8	322.0	72.3	3.9	208	205/2	319.4	70.5
11	8YF-43	36.959	116.216	7/7	310.1	70.2	2.7	518	0/0	310.1	70.2
	JR89-22	36.902	116.488	8/8	299.5	55.1	2.9	355	30/12	299.3	67.1
	reference mean			3/3			6.2	400		309.1	69.5
Tiva Canyon Member											
	7YF-1	37.167	116.223	8/8	171.7	-36.3	2.0	782	163/6	167.3	-36.9
	8YF-44	36.961	116.218	8/8	171.3	-43.0	2.1	718	0/0	171.3	-43.0
	9YF-35	36.994	116.224	8/8	168.6	-43.7	1.6	1256	205/2	167.1	-42.5
	JR87-4	36.906	116.487	11/11	163.9	-42.0	1.6	829	0/0	163.9	-42.0
	JR81-2	36.892	116.461	9/9	166.3	-39.7	2.5	420	22/3	168.4	-41.4
	M79-25	36.874	116.464	6/6	173.1	-39.6	3.1	474	76/6	173.7	-44.6
	M79-26	36.871	116.463	5/7	164.2	-37.8	4.4	302	73/5	164.4	-42.7
	M79-28	36.870	116.459	6/6	165.0	-41.6	2.7	625	73/5	168.0	-46.6
				Halfpint Range							
Ammonia Tanks Member											
23	8YF-10	36.959	115.932	8/8	36.4	60.5	3.9	202	170/38	326.5	65.1
Rainier Mesa Member											
12	7YF-12	37.119	116.007	5/7	172.9	-43.2	2.1	1288	165/23	151.4	-42.0
13	7YF-13	37.039	115.982	7/7	183.2	-57.8	2.1	804	170/5	175.2	-58.5
14	9YF-26	37.014	115.971	8/8	195.1	-43.0	1.3	1720	172/22	172.9	-47.7
15	9YF-25	37.015	115.971	8/8	197.4	-50.1	2.8	383	172/22	168.8	-54.5
17	7YF-9,10	36.984	115.981	19/19	187.1	-52.5	2.5	183	171/18	162.8	-53.8
				10/10†	193.1	-50.9	2.0	573	171/18	169.6	-54.4
18	9YF-39	36.974	115.987	8/8	187.3	-54.4	1.7	1098	184/17	164.4	-51.9
19	7YF-8	36.965	115.983	8/8	208.4	-54.6	1.9	868	170/21	175.5	-62.8
20	7YF-7	36.965	115.982	7/8	209.7	-48.1	4.2	209	172/28	172.4	-57.8
21	7YF-6	36.950	115.963	8/8	190.7	-53.2	1.9	827	170/25	155.4	-54.6
24	8YF-9	36.959	115.931	7/7	208.2	-47.8	2.4	625	170/38	165.4	-55.2
Topopah Spring Member											
16	9YF-27	36.987	115.977	7/7	331.1	77.6	2.5	598	179/18	294.5	63.9
22	9YF-38	36.950	115.962	6/7	303.0	72.3	2.4	752	179/19	286.2	55.0
25	8YF-8	36.959	115.928	8/8	47.2	76.3	2.0	784	170/33	279.5	67.4
26	8YF-11	36.932	115.908	7/7	54.9	85.9	3.7	261	167/23	261.7	70.7
27	FP	36.919	115.924	39/39	3.0	79.5	1.7	189	175/20	295.0	68.9

were found in basal parts of the Rainier Mesa Member in several stratigraphic traverses (Fig. 8). Compared to remanent directions from middle and upper parts of the Rainier Mesa, declinations in basal, pink, poorly welded parts of the sheet were shifted as much as 10°–15° counterclockwise at several sites (for example, 7YF-9 to 7YF-10, 8YF-54 in Fig. 8). The Rainier Mesa Member is believed to be composed of three individual flows (R. G. Warren, Los Alamos National Laboratory, 1990, oral commun.). Tentatively, the change of remanent directions observed in basal parts of the Rainier Mesa Member is ascribed to cooling of the lower flow of a compound cooling unit in a geomagnetic field of different direction than that recorded by overlying flows. The postulated boundary between the lower and middle flows is near the top of the poorly welded basal section of the Rainier Mesa Member and therefore is not readily apparent as a welding minimum. Empirically, however, the change of remanent directions commonly coincided with a color change from orangish pink to grayish purple at several sites (Fig. 8). A declination shift of opposite sense and smaller magnitude may also occur between a zone of samples just above the poorly welded basal section and higher parts of the tuff at sites 7YF-16 and 8YF-54 (Fig. 8). This change, however, does not correlate with any previously identified flow boundaries. For the analysis of vertical-axis rotation discussed in the following section, site means were calculated that omitted samples suspected to come from the basal flow (Table 1).

Estimates of Vertical-Axis Rotation

If the pattern of arcuate structures in the FPMM area arose from the proposed clockwise flexure of Carr (1984), site means of tuffs from this area should have declinations that are as

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TABLE 1. (*Continued*)

No.	Id.	Lat.	Long.	N/N_0	D_0	I_0	α_{95}	k	St/Dp	D_1	I_1
				French Peak–Massachusetts Mountain area							
Ammonia Tanks Member											
40	8YF-34	36.903	115.981	8/8	356.5	34.8	3.0	337	85/27	357.6	61.8
41	8YF-24	36.896	115.980	7/7	332.8	24.4	2.8	479	45/33	345.0	54.4
44	JR87-16	36.895	115.988	4/8	298.4	40.8	4.8	375	340/24	321.1	53.3
46	8YF-25	36.895	115.972	8/8	358.6	42.2	6.0	87	58/21	14.8	58.8
50	8YF-35	36.886	115.989	9/9	329.0	31.5	8.6	37	50/25	347.2	55.6
51	8YF-22	36.878	115.980	6/6	326.8	40.2	4.5	225	42/26	339.8	64.5
54	7YF-18	36.861	115.992	4/7	345.4	43.7	4.4	432	45/12	353.1	53.7
56	7YF-15	36.926	116.032	7/7	2.8	47.1	3.1	393	135/12	351.7	55.2
Rainier Mesa Member											
28	8YF-17	36.916	115.952	7/7	150.4	-40.8	4.7	164	37/10	154.8	49.8
29	8YF-18	36.914	115.956	8/8	170.9	37.3	2.1	684	55/20	181.8	54.5
30	8YF-16	36.913	115.948	5/7	147.1	19.3	3.3	547	70/22	143.9	40.6
33	8YF-14	36.903	115.941	5/7	161.5	-30.7	2.1	1340	110/4	157.0	-41.0
35	8YF-12	36.894	115.937	14/17	166.6	-30.9	4.1	95	71/11	167.5	41.8
				6/9†	172.0	-25.7	5.0	181	71/11	173.4	36.5
37	8YF-31	36.912	115.992	7/8	190.8	-10.3	4.9	180	100/33	189.1	-45.2
39	8YF-33	36.903	115.981	12/13	176.9	-27.3	6.6	45	85/25	177.7	52.3
				9/10†	183.0	-29.3	4.4	137	85/25	186.9	-53.9
43	JR87-17	36.895	115.990	4/8	123.6	36.3	3.9	544	340/26	145.3	47.9
45	8YF-23	36.894	115.982	7/8	138.7	33.6	1.9	986	65/15	140.2	48.5
47	8YF-26	36.894	115.966	8/8	179.1	25.1	1.4	1471	60/26	190.0	46.7
49	8YF-36	36.886	115.990	6/8	140.2	-33.6	4.1	265	19/29	159.6	56.8
52	8YF-21	36.877	115.979	8/8	167.3	-35.9	2.7	409	40/23	183.4	52.3
53	7YF-19	36.870	115.988	8/8	164.8	-28.0	2.9	380	30/24	178.5	-43.1
55	7YF-17	36.861	115.993	10/10	169.5	-42.7	2.1	558	45/12	177.7	52.1
57	7YF-14	36.926	116.031	12/13	181.2	-42.9	1.5	850	135/12	171.5	-50.9
Topopah Spring Member											
31	8YF-19	36.911	115.955	8/8	320.8	51.3	3.1	321	85/16	303.7	63.2
32	8YF-20	36.908	115.951	8/8	321.7	52.3	2.7	421	110/17	298.3	58.3
34	8YF-13	36.902	115.940	5/7	314.6	61.4	2.9	719	100/12	291.6	66.3
36	8YF-30	36.912	115.994	8/8	0.0	44.4	3.3	280	100/25	326.6	75.7
38	8YF-32	36.904	115.981	6/8	336.3	36.0	4.0	280	95/34	310.0	61.6
42	JR87-18	36.895	115.991	7/8	264.4	43.0	3.4	321	340/26	274.7	73.5
48	8YF-37	36.887	115.992	8/8	287.8	33.3	2.6	465	50/30	277.8	62.5

Note: No., number of site in Figures 9 and 10; Id., site identification; Lat., Long., latitude (°N) and longitude (°W) of site; N/N_0, number of samples accepted/number of samples demagnetized; D_0, I_0, *in situ* site-mean declination (degrees clockwise from north) and inclination (degrees from horizontal, negative values are up); α_{95}, cone of 95% statistical confidence about site mean; k, concentration parameter (Fisher, 1953); St/Dp, strike and dip of site (dip 90° clockwise from strike); D_1, I_1, tilt-corrected declination and inclination of site mean.

*Mean derived from remagnetization circle analysis–α_{95} is estimated as average of maximum and minimum axes of confidence ellipse.

†Site mean calculated by omitting samples from basal part of Rainier Mesa Member.

much as 90° clockwise from declinations of site means from the Halfpint Range. Tilt-corrected site means for each unit (Fig. 6) illustrate that although site means from the FPMM area are more dispersed than those from the other areas, their declinations are not systematically clockwise from those from the Halfpint Range or the mesa areas. Therefore, the tuffs in the FPMM area have not been strongly rotated clockwise about a common vertical or steeply inclined axis.

Apparent vertical-axis rotations of the individual sites are quantified by comparison to reference mean directions (Table 1) for the three tuffs. The reference directions are averages of sites from Rainier Mesa, Aqueduct Mesa, Pahute Mesa, Shoshone Mountain, and northern Yucca Mountain areas (Figs. 1 and 2). Tilt-corrected site means from each of the three tuffs in the mesa areas (Fig. 6) show low dispersion (k values for the reference means range from 400 to 981, Table 1), indicating that no significant vertical-axis rotation occurred between the different mesa sites. Although the Tiva Canyon Member (12.7 Ma; Sawyer and others, 1990) of the Paintbrush Tuff is not present in the FPMM area and the Halfpint Range, the similarity of directions from new and previously published site means from this well-studied unit (Rosenbaum, 1986; Rosenbaum and others, 1991) from Rainier Mesa, Shoshone Mountain, and northern Yucca Mountain areas (Fig. 6; Table 1) is independent evidence of insignificant vertical-axis rotation between the mesa areas. Due to their steep inclinations, site means used to derive the reference direction for the Topopah Spring Member have a 20° range of declination (Table 1) despite being relatively well grouped (k = 400).

Angular differences between declinations of individual site means and reference directions for each unit reflect the vertical-axis rotation of each site as well as dispersion that might arise from other sources (for example, tilt-correction errors). Vertical-axis rotations inferred from these declination differences range from +23° to −31° for the Ammonia Tanks Member (positive values are clockwise), +19° to −31° for the Rainier Mesa Member, and +18° to −47° for the Topopah Spring Member (Figs. 9 and 10). There is no apparent correlation between amount and sense of declination difference for individual sites and the strike of structures in the FPMM area (Fig. 10).

Although angular discordances between individual site means and the reference directions are too small and commonly of incorrect sense (Figs. 9 and 10) to produce the pattern of arcuate structures in the FPMM area by the flexure model, they merit attention. Three factors probably increase dispersion of paleomagnetic data from the FPMM area and the Halfpint Range relative to data from the mesa areas: (1) tilt-correction errors, (2) initial intraunit dispersion of remanent directions, and (3) vertical- or inclined-axis rotation of local scale.

As other workers have noted (Hagstrum and Gans, 1989; Rosenbaum and others, 1991), significant dispersion of site means from ash-flow

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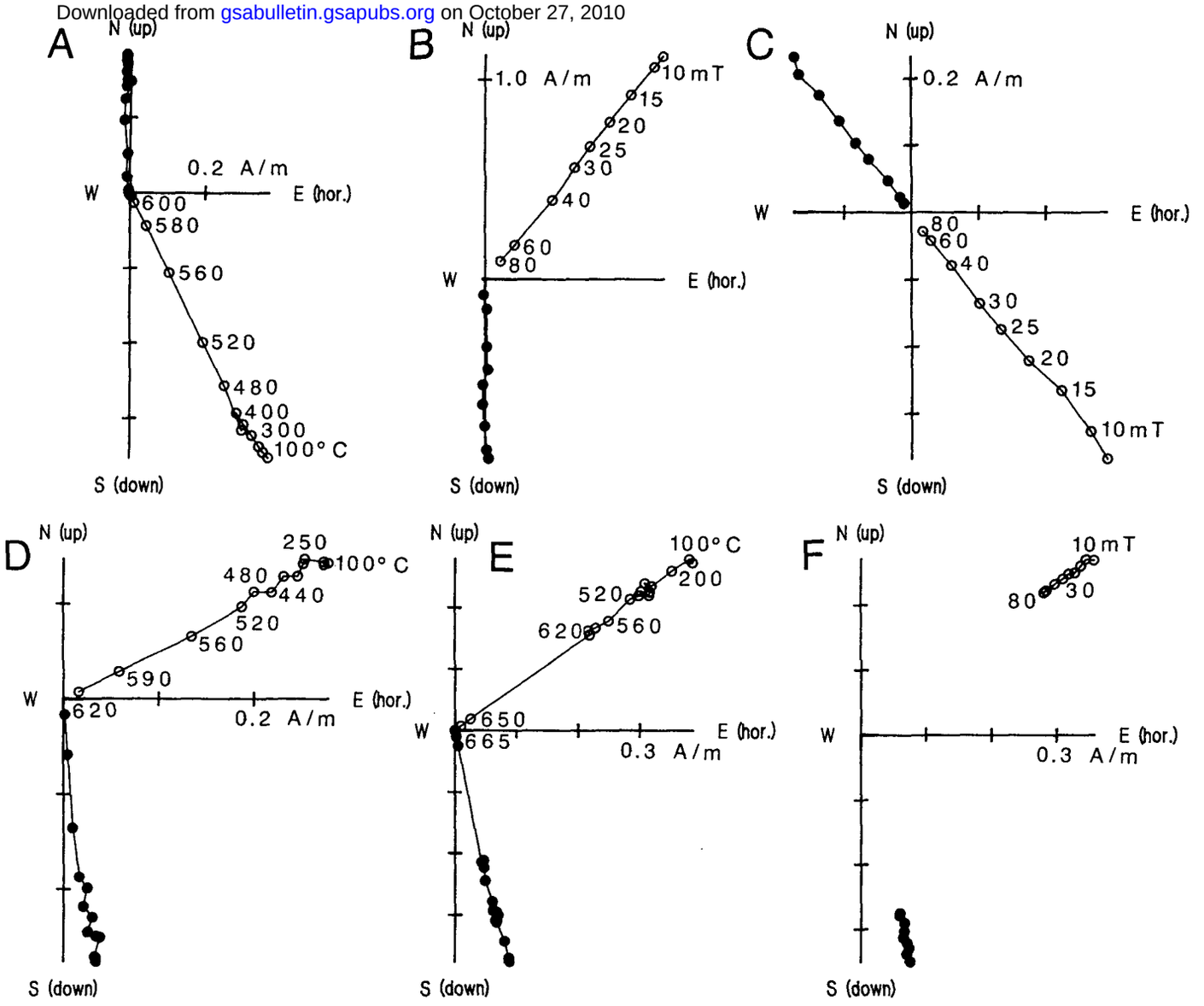


Figure 5. Representative orthogonal demagnetization diagrams, uncorrected for tectonic tilt, from Miocene ash-flow tuffs. Solid and open circles are projections on horizontal and vertical planes, respectively. Demagnetization levels are given in milliTesla (mT) or degrees Celsius (°C). A. Ammonia Tanks Member, site 7YF-3. B. Rainier Mesa Member, site 7YF-9. C. Topopah Spring Member, site 8YF-19. D. Rainier Mesa Member, site 8YF-12. E. Rainier Mesa Member, site 7YF-17, sample 1B. F. Rainier Mesa Member, site 7YF-17, sample 1A.

tuffs may arise from errors in tilt corrections. Tilt-correction errors may be greater for sites from the Halfpint Range and FPMM area for two reasons. In the FPMM area and the Halfpint Range, the Rainier Mesa and Ammonia Tanks Members are generally only poorly to moderately welded. Due to weak welding, compaction was not great, and any initial fluctuation of pumice alignment from paleohorizontal (for example, related to mantled paleotopography or flow-related imbrication; Elston and Smith, 1970) may persist. Because the Halfpint Range and FPMM area commonly are broken by closely spaced faults (both mapped faults and unmapped small-displacement faults), it generally was not accurate to estimate attitudes from the trace of contacts over topography. Such regional attitudes, which were used for some sites from the mesa areas, are less likely to be affected by local irregularities than compaction foliations. Errors in declination arising from inaccurate tilt corrections are potentially greatest for rocks with steeply inclined magnetization because small errors can produce relatively large changes in declination. This may partly explain why declinations of steeply inclined site means for the Topopah Spring Member show greatest dispersion.

As previously noted, welding postdating remanence acquisition in the Topopah Spring Member or limited paleosecular variation recorded by different flows of the Rainier Mesa Member could produce intraunit dispersion and contribute to declination differences for some sites. Although samples from the basal Rainier Mesa Member were omitted from site-mean calculations where possible, samples from two sites from the Halfpint Range (7YF-6, 7YF-7) came only from the basal, pink, poorly welded zone, and therefore their means could not be recalculated in this manner. Intraunit dispersion within the Rainier Mesa Member might produce declination differences on the order of 10° for individual, stratigraphically restricted sites.

Finally, for some individual fault blocks, the discordance of declinations is consistent among several sites from different units (for example, sites 36 and 37, or sites 42, 43, and 44 in Fig.

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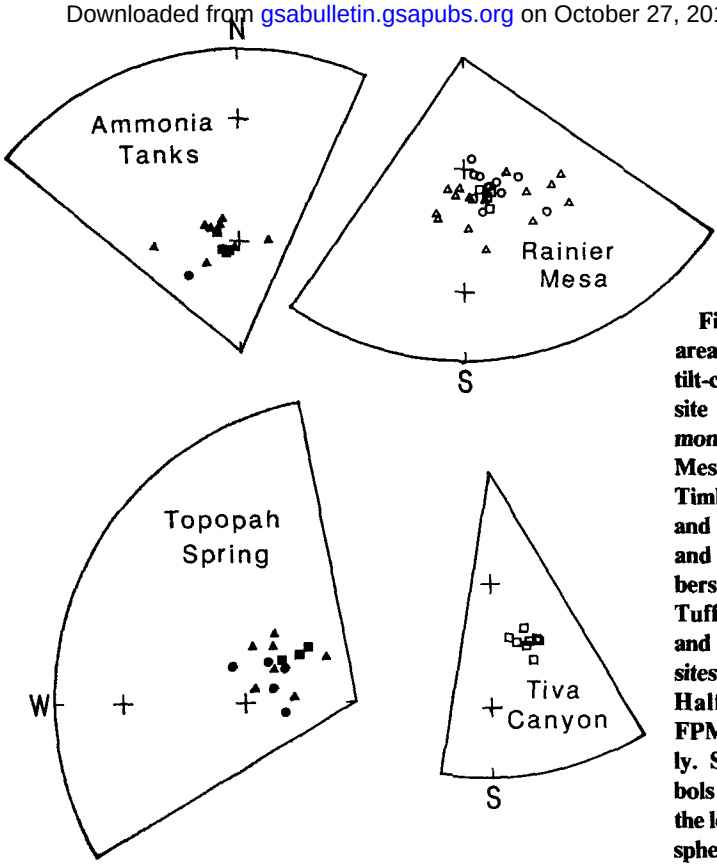


Figure 6. Partial equal-area projections showing tilt-corrected directions of site means from the Ammonia Tanks and Rainier Mesa Members of the Timber Mountain Tuff and the Topopah Spring and Tiva Canyon Members of the Paintbrush Tuff. Squares, circles, and triangles are from sites from the mesa areas, Halfpint Range, and FPMM area, respectively. Solid and open symbols are projections on the lower and upper hemispheres, respectively.

10) and the discordances seem too great to be explained solely by tilt-correction errors or intraunit dispersion. These blocks were likely affected by true vertical- or inclined-axis rotation. Such rotations, however, appear to be local in extent and to be of both clockwise and counterclockwise sense, rather than the systematic clockwise rotation predicted by the flexure model.

TECTONIC IMPLICATIONS

Lack of systematic clockwise rotation of rocks in the FPMM area disproves the flexure postulated for the Yucca-Frenchman shear zone. Coupled with the rarity of northwest-striking faults in the FPMM area that could accommodate right-lateral offset, the paleomagnetic data suggest that little right-lateral shear strain could have occurred along the postulated Yucca-Frenchman shear zone after deposition of the middle Miocene volcanic rocks. Paleomagnetic data from this study highlight the general difficulty of detecting vertical-axis rotation from geologic map patterns alone.

Lack of systematic vertical-axis rotation in the FPMM area indicates that faults within Miocene volcanic rocks originated with arcuate trends. The potential causes for this arcuate nucleation

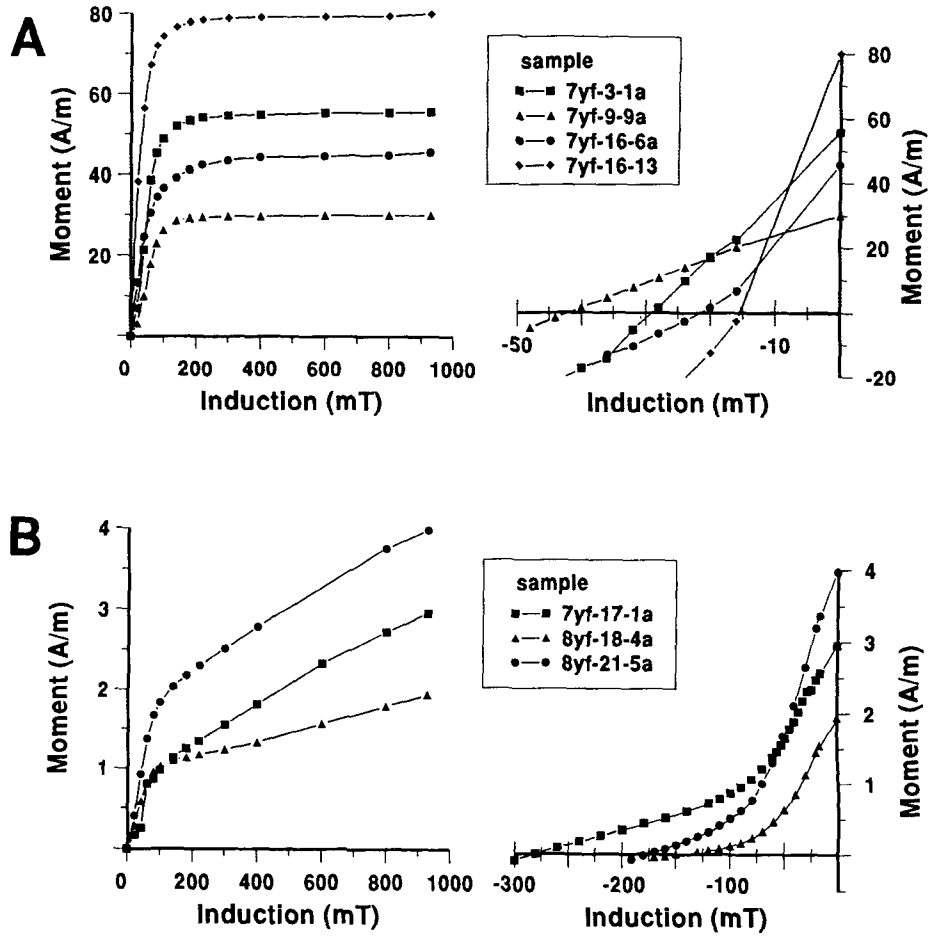


Figure 7. Diagrams depicting acquisition of isothermal remanent magnetization (left) and direct-field demagnetization (right) showing representative behavior for the ash-flow tuff samples. Inductions are in milliTesla (mT). A. Samples from the Ammonia Tanks and Rainier Mesa Members whose behavior suggests that their magnetic mineralogy is dominated by magnetite or maghemite. B. Samples from a vapor-phase zone of the upper Rainier Mesa Member from the FPMM area whose behavior indicates the presence of significant hematite in addition to magnetite.

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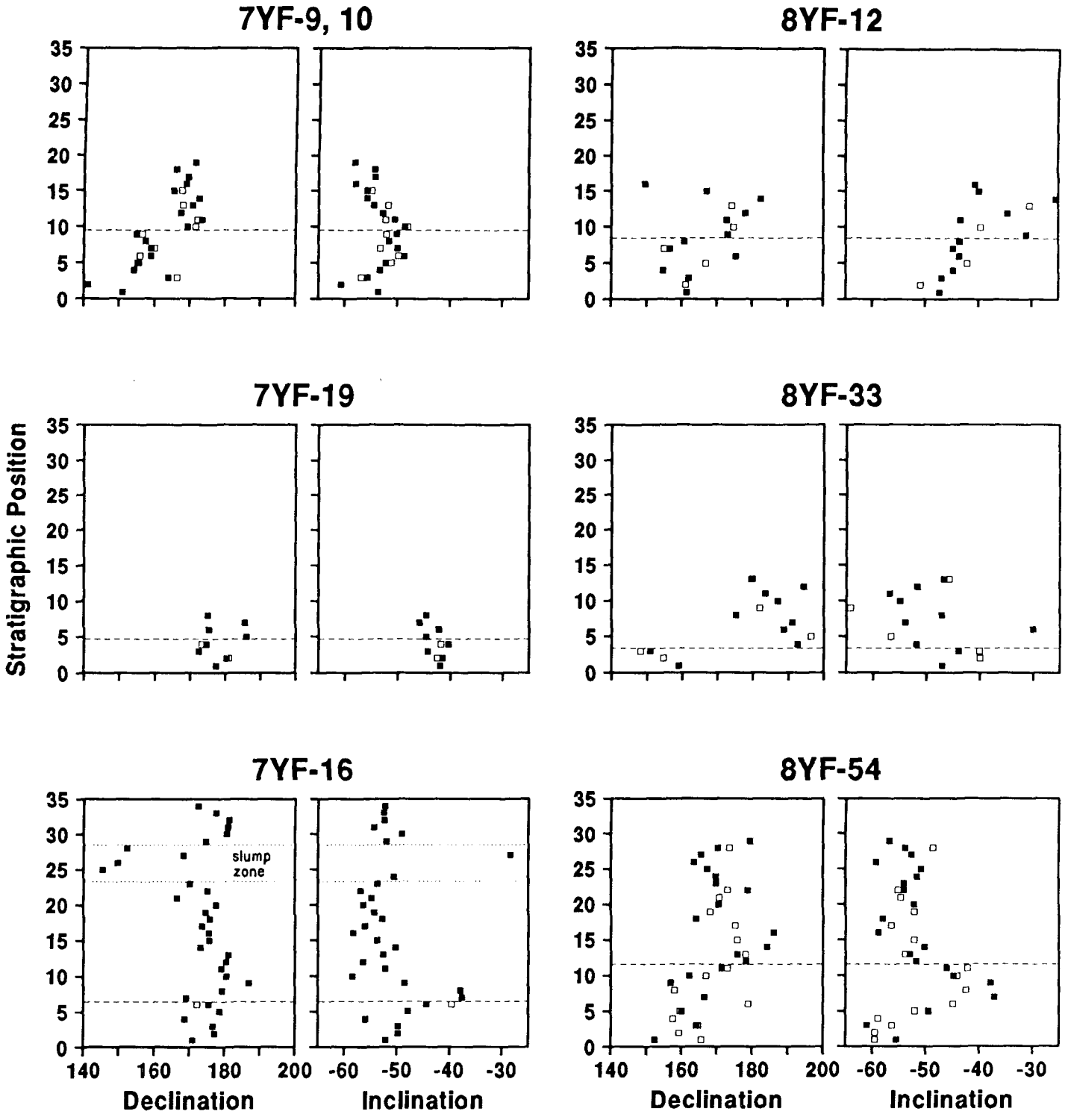


Figure 8. Plots of declination and inclination for individual samples at sites in the Rainier Mesa Member that were collected as traverses across an extended stratigraphic interval of the ash-flow sheet. Only relative stratigraphic positions of individual samples are shown, but samples were typically separated by two to five meters. Dashed line shows approximate top of basal, poorly to moderately welded, pink to orange part of the sheet. Solid and open squares show samples treated by alternating-field and thermal demagnetization, respectively. A counterclockwise shift of declinations that is seen in the basal part of the sheet at some sites (for example, 7YF-9 to 7YF-10) is tentatively attributed to cooling of a basal flow of the compound cooling unit in a geomagnetic field of different direction than that recorded by the overlying part of the sheet.

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are of tectonic interest. Two viable explanations are (1) a control by prevolcanic structures, or (2) a stress reorientation within the accommodation zone.

Structures within the Walker Lane belt have had protracted histories, and many may have originated during Mesozoic time (Stewart, 1988). Therefore, if the Yucca-Frenchman shear zone was a prevolcanic structure, it is conceivable that it, or an associated orocline, was reactivated and controlled the formation of arcuate faults during postvolcanic extension in the FPMM area. Because clockwise flexure in the FPMM area was cited as the principal evidence for the Yucca-Frenchman shear zone (Carr, 1984), however, the existence of this zone must be questioned. Other evidence cited for the postulated shear zone is not unique. A curvature of Bouguer gravity highs that was inferred to reflect right-lateral offset of Paleozoic rocks beneath the FPMM area (Carr, 1984) more simply marks the northwestern and southeastern margins of the Frenchman Flat and Yucca Flat basins, respectively. Paleozoic rocks in exposures along the trend of the postulated shear zone to the northwest or southeast are not offset by right-lateral faults. There is an apparent termination of seismicity within central and southwestern Frenchman Flat along a northwest-trending line parallel to the postulated Yucca-Frenchman shear zone (Rogers and others, 1983), but there is no strong alignment of epicenters along this zone. Therefore, although not precluded, the possibility that structures related to a prevolcanic Yucca-Frenchman shear zone controlled

TABLE 2. FOLD TESTS FOR MIOCENE ASH-FLOW TUFFS

Concentration parameter type	N	k uncorrected	k corrected	k ratio	Positive test?, confidence level
		Ammonia Tanks Member			
Normal	12	15	101	6.7	Yes, >99%
Inclination only	12	17	171	9.9	Yes, >99%
		Rainier Mesa Member			
Normal	32	19	70	3.8	Yes, >99%
Inclination only	32	24	110	4.6	Yes, >99%
		Topopah Spring Member			
Normal	15	13	92	7.1	Yes, >99%
Inclination only	15	10	102	10.3	Yes, >99%

Note: N, number of site means used in calculation.

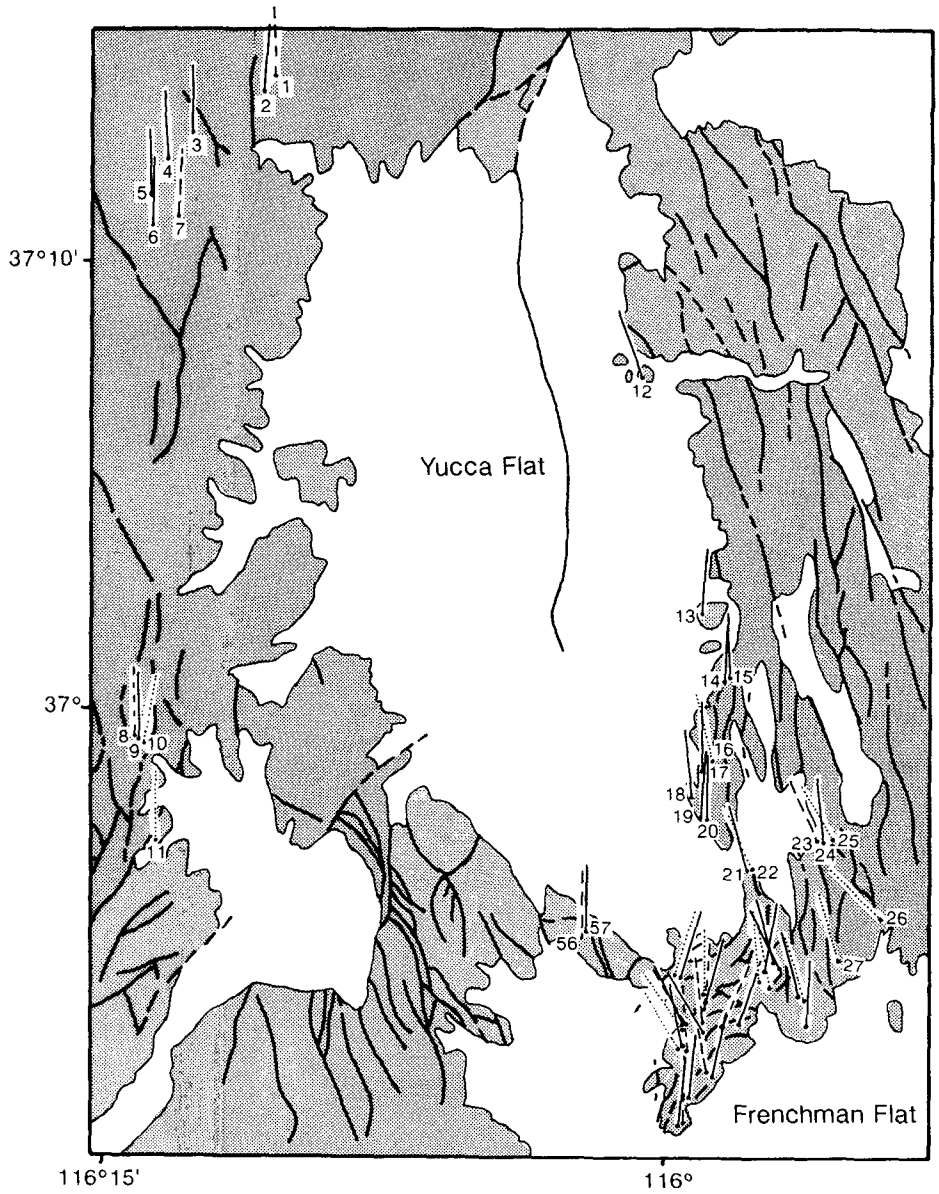


Figure 9. Estimates of vertical-axis rotation (angle between line and geographic north) for individual sites (dots). Amount of rotation is the angular difference between the declination of an individual site mean and the declination for the reference mean for each unit (Table 1). Numbers by site locations are keyed to individual sites on Table 1. Numbers for closely spaced sites in the FPMM area southeast of Yucca Flat are shown in Figure 10. Symbols for rotation estimates from the different units are: dashed lines, Ammonia Tanks Member; solid lines, Rainier Mesa Member; dotted lines, Topopah Spring Member. Heavy lines are faults.

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subsequent nucleation of arcuate faults in the FPMM area is not favored because there is little direct evidence for existence of the postulated shear zone.

The origin of arcuate strikes of faults in the FPMM area is most likely related to the location within an extensional accommodation zone. Arcuate faults are common within accommodation zones of other extended areas (Bosworth, 1985; Rosendahl and others, 1986; Ebinger, 1989; Coffield and Schamel, 1989; Faulds and others, 1990), and they have also been produced in clay deformation models (Serra and Nelson, 1988). We propose that arcuate nucleation of faults in the FPMM area may have originated from a local reorientation of the tectonic stress field within the accommodation zone. One explanation, an analogy to propagating crack tips (Pollard and others, 1982), would attribute a local stress reorientation to a region of overlap between two fault zones that simultaneously propagated into the accommodation zone from the north and south (Fig. 11). The pattern of arcuate faults in the FPMM area is geometrically similar to curved terminations observed in other tensile structures, such as joints (Pollard and others, 1982; Segall and Pollard, 1983), dikes (Delaney and Pollard, 1981), and oceanic spreading centers (MacDonald and Fox, 1983; Pollard and Aydin, 1984). Aydin and Pollard (1982) attributed irregular forms of some normal faults to such stress-field interactions. The analogy between propagating fault zones and crack tips, however, may be only approximate. The observation that several faults curve clockwise in a parallel manner in the FPMM area (Fig. 3) suggests that extension may have been distributed within zones that encompassed several faults and, at least at shallow crustal levels, that were several kilometers wide rather than being confined to single faults propagating in undeformed rock as schematically depicted in Figure 11. Nevertheless, if strain rates across such nucleating zones exceeded those of surrounding regions, stress reorientation at an overlap within the accommodation zone might still be expected. If this model is valid, the northern and southern interacting faults zones would encompass the southwestern edge of the Halfpint Range and all of southern Massachusetts Mountain, respectively, in the present geography. In this type of model for stress reorientation, the accommodation zone was important because it provided an area where propagating fault zones could simultaneously overlap, thereby allowing local reorientation of the stress field.

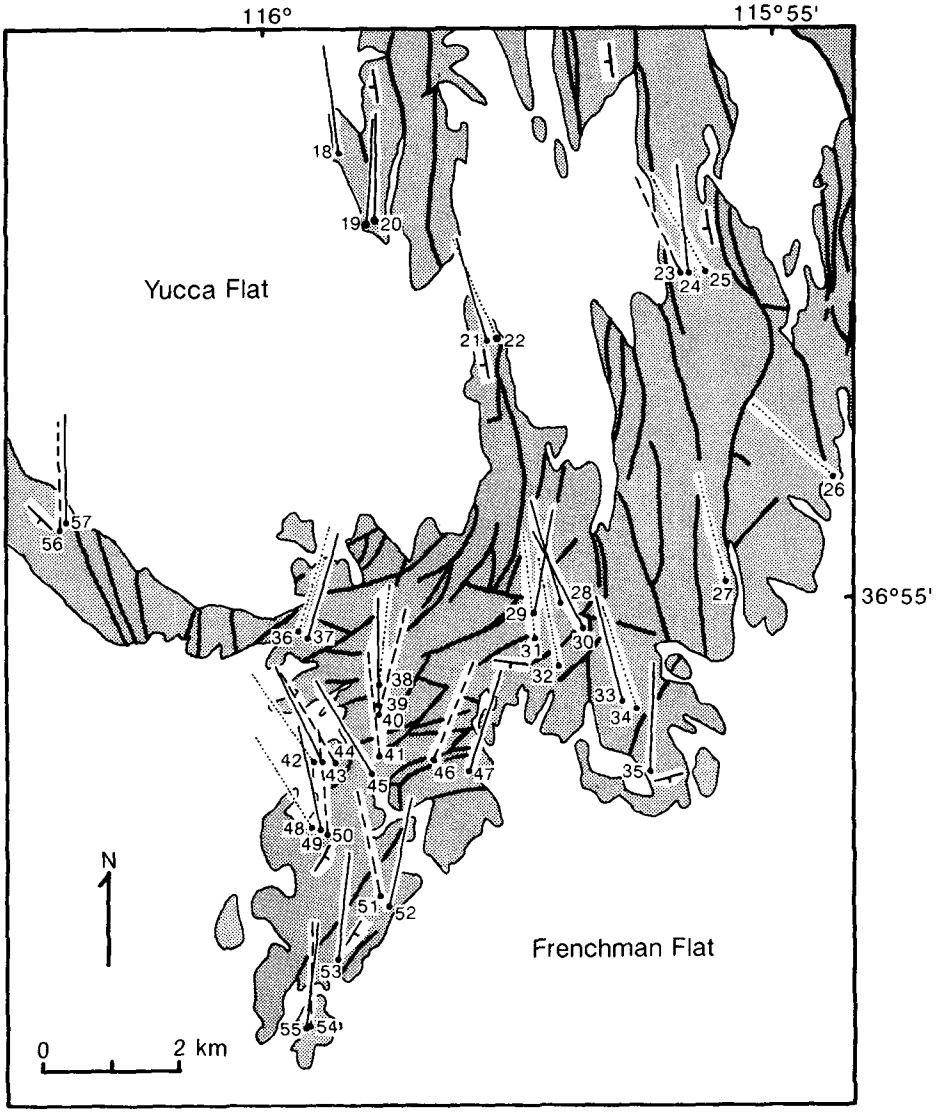


Figure 10. Estimates of vertical-axis rotation (angle between line and geographic north) for individual sites (dots) from the FPMM and southern Halfpint Range areas. Amount of rotation is the angular difference between the declination of an individual site mean and the declination for the reference mean for each unit (Table 1). Numbers by site locations are keyed to individual sites on Table 1. Symbols for rotation estimates from the different units are: dashed lines, Ammonia Tanks Member; solid lines, Rainier Mesa Member; dotted lines, Topopah Spring Member. Heavy lines are faults. The lack of systematic clockwise vertical-axis rotation in the FPMM area indicates that the arcuate structures in the area did not form by clockwise flexure above the postulated Yucca-Frenchman shear zone.

Paleomagnetic data from this study also provide constraint on the structural development of the Yucca Flat basin. Cole (1987), considering that the depth of the southern part of Yucca Flat basin is greater than the northern part, suggested that basin formation may have occurred with some component of counterclockwise rotation of the Halfpint Range about a pole north of Yucca Flat. The mean of eight sites, however, from the Rainier Mesa Member in the Halfpint Range is D = 167.1°, I = –53.6°, α_{95} = 5.5° (omitting sites 7YF-6 and 7YF-7, suspected to be within the basal flow), indicating an insignifi-

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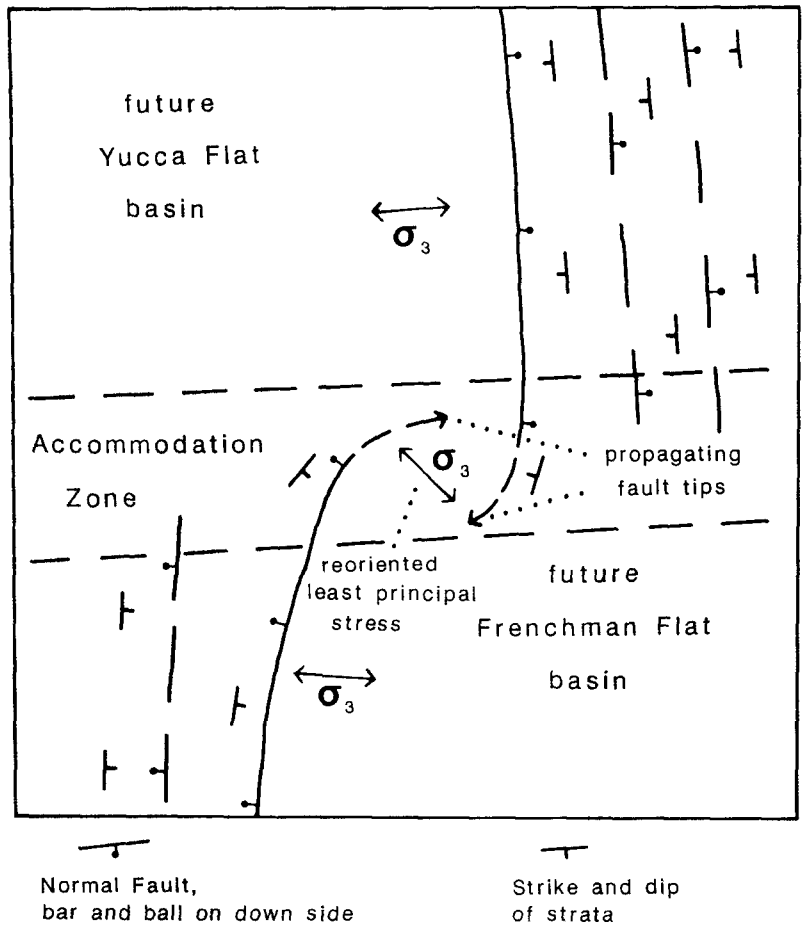


Figure 11. Schematic paleogeographic diagram depicting development of arcuate structures in the FPMM area as the result of local reorientation of the stress field between overlapping segments of two fault zones that simultaneously propagated into an accommodation zone.

cant counterclockwise rotation of –3.0° ± 7.9° (Demarest, 1983) when compared to the reference direction from the mesa areas. If it is assumed that the Halfpint Range moved as a coherent mass during opening of the Yucca Flat basin, the lack of significant pivotal motion implies that total extension across the basin should be approximately constant along its length. Deepening of the southern part of the basin could have been controlled by other factors such as focusing of slip onto fewer faults or a steepening of fault dips.

CONCLUSIONS

Paleomagnetic data obtained from the Miocene Ammonia Tanks and Rainier Mesa Members of the Timber Mountain Tuff and the Topopah Spring Member of the Paintbrush Tuff indicate that no regionally significant vertical-axis rotation affected the French Peak–Massachusetts Mountain area. Therefore, the observed pattern of arcuate structures is not due to clockwise flexure above the postulated Yucca-Frenchman right-lateral shear zone of Carr (1984). The pattern of arcuate structures in the French Peak–Massachusetts Mountain area may reflect local reorientation of the stress field within an accommodation zone that might have occurred when two fault zones simultaneously propagated into the zone from separate tilt-block domains. Paleomagnetic data from this study highlight the general difficulty of detecting vertical-axis rotation from geologic map patterns alone.

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REFERENCES CITED

Anderson, R. E., 1973, Large-magnitude late Tertiary strike-slip faulting north of Lake Mead, Nevada: U.S. Geological Survey Professional Paper 794, 18 p.

Aydin, A., and Pollard, D. D., 1982, Origin of zig-zag pattern of normal faults: Geological Society of America Abstracts with Programs, v. 14, p. 436.

Barnes, H., Ekren, E. B., Rodgers, C. L., and Hedlund, D. C., 1982, Geologic and tectonic maps of the Mercury Quadrangle, Nye and Clark counties, Nevada: U.S. Geological Survey Miscellaneous Investigations Map I-1197, scale 1:24,000.

Bosworth, W., 1985, Geometry of propagating continental rifts: Nature, v. 316, p. 625–627.

Bosworth, W., Lambiase, J. J., and Keisler, R., 1986, A new look at Gregory's rift: The structural style of continental rifting: Eos (American Geophysical Union Transactions), v. 67, p. 577–583.

Burchfiel, B. C., 1965, Structural geology of the Specter Range quadrangle, Nevada, and its regional significance: Geological Society of America Bulletin, v. 76, p. 175–192.

Byers, F. M., Jr., and Barnes, H., 1967, Geologic map of the Paiute Ridge quadrangle, Nye and Lincoln Counties, Nevada: U.S. Geological Survey Geologic Quadrangle Map GQ-577, scale 1:24,000.

Byers, F. M., Jr., Carr, W. J., Orkild, P. P., Quinlivan, W. D., and Sargent, K. A., 1976, Volcanic suites and related calderas of Timber Mountain–Oasis Valley caldera complex, southern Nevada: U.S. Geological Survey Professional Paper 919, 70 p.

Byers, F. M., Jr., Carr, W. J., and Orkild, P. P., 1989, Volcanic centers of southwestern Nevada: Evolution of understanding, 1960–1988: Journal of Geophysical Research, v. 94, p. 5908–5924.

Carr, W. J., 1974, Summary of tectonic and structural evidence for stress orientation at the Nevada Test Site: U.S. Geological Survey Open-File Report 74-176, 53 p.

——1984, Regional structural setting of Yucca Mountain, southwestern Nevada, and late Cenozoic rates of tectonic activity in part of the southwestern Great Basin, Nevada and California: U.S. Geological Survey Open-File Report 84-854, 109 p.

Coffield, D. Q., and Schamel, S. S., 1989, Surface expression of an accommodation zone within the Gulf of Suez rift, Egypt: Geology, v. 17, p. 76–79.

Cole, J. C., 1987, Breakup of an allochthonous slab: Origin of the Yucca Flat Basin, Nevada Test Site: Geological Society of America Abstracts with Programs, v. 19, p. 623.

Crowe, B. M., Vaniman, D. T., and Carr, W. J., 1983, Status of volcanic hazard studies for the Nevada Nuclear Waste Storage Investigations: Los Alamos National Laboratory Report LA-9325-MS, 47 p.

Dalrymple, G. B., Cox, A., and Doell, R. R., 1965, Potassium-argon age and paleomagnetism of the Bishop Tuff, California: Geological Society of America Bulletin, v. 76, p. 665–673.

Delaney, P. T., and Pollard, D. D., 1981, Deformation of host rocks and flow of magma during growth of minette dikes and breccia-bearing intrusions near Shiprock, New Mexico: U.S. Geological Survey Professional Paper 1201, 61 p.

Demarest, H. H., 1983, Error analysis for the determination of tectonic rotation from paleomagnetic data: Journal of Geophysical Research, v. 88, p. 4321–4328.

Ebinger, C. J., 1989, Geometric and kinematic development of border faults and accommodation zones, Kivu-Rusizi rift, Africa: Tectonics, v. 8, p. 117–133.

Ekren, E. B., 1968, Geologic setting of Nevada Test Site and Nellis Air Force Range, *in* Eckel, E. B., ed., Nevada Test Site: Geological Society of America Memoir 110, p. 11–19.

Elston, W. E., and Smith, E. I., 1970, Determination of flow direction of rhyolitic ash-flow tuffs from fluidal textures: Geological Society of America Bulletin, v. 81, p. 3393–3406.

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Faulds, J. E., Hillmeyer, F. L., and Smith, E. I., 1988, Geometry and kinematics of a Miocene "accommodation zone" in the central Black and southern Eldorado Mountains, Arizona and Nevada, *in* Weide, D. L., and Faber, M. L., eds., This extended land, geological journeys in the southern Basin and Range: Geological Society of America Field Trip Guidebook, Cordilleran Section, p. 293–310.

Faulds, J. E., Geissman, J. W., and Mawer, C. K., 1990, Structural development of a major extensional accommodation zone in the Basin and Range province, northwestern Arizona and southern Nevada; Implications for kinematic models of continental extension, *in* Wernicke, B. P., ed., Basin and Range extensional tectonics near the latitude of Las Vegas, Nevada: Geological Society of America Memoir 176, p. 37–76.

Fisher, R. A., 1953, Dispersion on a sphere: Royal Society of London Proceedings, v. A217, p. 295–305.

Geissman, J. W., 1980, Paleomagnetism of ash-flow tuffs: Microanalytical recognition of TRM components: Journal of Geophysical Research, v. 85, p. 1487–1499.

Gose, W. A., 1970, Palaeomagnetic studies of Miocene ignimbrites from Nevada: Royal Astronomical Society Geophysical Journal, v. 20, p. 241–252.

Gromme, C. S., McKee, E. H., and Blake, M. C., 1972, Paleomagnetic correlations and potassium-argon dating of middle Tertiary ash-flow sheets in the eastern Great Basin, Nevada and Utah: Geological Society of America Bulletin, v. 83, p. 1619–1638.

Hagstrum, J. T., and Gans, P. B., 1989, Paleomagnetism of the Oligocene Kalamazoo Tuff: Implications for middle Tertiary extension in east central Nevada: Journal of Geophysical Research, v. 94, p. 1827–1842.

Halls, H. C., 1976, A least-squares method to find a remanence direction from converging remagnetization circles: Royal Astronomical Society Geophysical Journal, v. 45, p. 297–304.

Hinrichs, E. N., and McKay, E. J., 1965, Geologic map of the Plutonium Valley quadrangle, Nye and Lincoln Counties, Nevada: U.S. Geological Survey Geologic Quadrangle Map GQ-384, scale 1:24,000.

Hudson, M. R., and Rosenbaum, J. G., 1990, Paleomagnetism of the southwestern Nevada volcanic field: Absence of vertical-axis rotation: Eos (American Geophysical Union Transactions), v. 71, p. 1295.

Kirschvink, J. L., 1980, The least-squares line and plane and analysis of paleomagnetic data: Royal Astronomical Society Geophysical Journal, v. 62, p. 699–718.

Longwell, C. R., 1974, Measure and date of movement on Las Vegas Valley shear zone, Clark County, Nevada: Geological Society of America Bulletin, v. 85, p. 985–990.

MacDonald, K. C., and Fox, P. J., 1983, Overlapping spreading centers: New accretion geometry on the East Pacific Rise: Nature, v. 302, p. 55–58.

MacDonald, W. D., 1980, Net tectonic rotation, apparent tectonic rotation, and the structural tilt correction in paleomagnetism studies: Journal of Geophysical Research, v. 85, p. 3659–3669.

Magill, J. R., Wells, R. E., Simpson, R. W., and Cox, A. V., 1982, Post 12 m.y. rotation of southwestern Washington: Journal of Geophysical Research, v. 87, p. 3761–3776.

McElhinny, M. W., 1964, Statistical significance of the fold test in palaeomagnetism: Royal Astronomical Society Geophysical Journal, v. 89, p. 338–340.

McFadden, P. L., and Reid, A. B., 1982, Analysis of paleomagnetic inclination data: Royal Astronomical Society Geophysical Journal, v. 69, p. 307–319.

McIntosh, W. C., 1991, Evaluation of paleomagnetism as a correlation criterion for Mogollon-Datil ignimbrites, southwestern New Mexico: Journal of Geophysical Research, v. 96, p. 13459–13483.

McKeown, F. A., Healy, D. L., and Miller, C. H., 1976, Geologic map of the Yucca Lake quadrangle, Nye County, Nevada: U.S. Geological Survey Geologic Quadrangle Map GQ-1327, scale 1:24,000.

Nelson, M. R., and Jones, C. H., 1987, Paleomagnetism and crustal rotations along a shear zone, Las Vegas Range, southern Nevada: Tectonics, v. 6, p. 13–33.

Onstott, T. C., 1980, Application of the Bingham distribution function in paleomagnetic studies: Journal of Geophysical Research, v. 85, p. 1500–1510.

Pollard, D. D., and Aydin, A., 1984, Propagation and linkage of oceanic ridge segments: Journal of Geophysical Research, v. 89, p. 10,017–10,028.

Pollard, D. D., Segall, P., and Delaney, P. T., 1982, Formation and interpretation of dilatant echelon cracks: Geological Society of America Bulletin, v. 93, p. 1291–1303.

Poole, F. G., 1965, Geologic map of the Frenchman Flat quadrangle, Nye, Lincoln, and Clark counties, Nevada: U.S. Geological Survey Geologic Quadrangle Map GQ-456, scale 1:24,000.

Poole, F. G., 1974, Flysch deposits of the Antler foreland basin, western United States, *in* Dickinson, W. R., ed., Tectonics and sedimentation: Society of Economic Paleontologists and Mineralogists Special Publication 22, p. 58–82.

Poole, F. G., Elston, D. P., and Carr, W. J., 1965, Geologic map of the Cane Springs quadrangle, Nye County, Nevada: U.S. Geological Survey Geologic Quadrangle Map GQ-455, scale 1:24,000.

Reches, Z., 1983, Analysis of faulting in three-dimensional strain field: Tectonophysics, v. 47, p. 109–129.

Reynolds, R. L., 1977, Paleomagnetism of tuffs of the Yellowstone Group: Journal of Geophysical Research, v. 82, p. 3677–3693.

Rogers, A. M., Harmsen, S. C., Carr, W. J., and Spence, W., 1983, Southern Great Basin seismological data report for 1981 and preliminary data analysis: U.S. Geological Survey Open-File Report 83-669, 240 p.

Rosenbaum, J. G., 1986, Paleomagnetic directional dispersion produced by plastic deformation in a thick Miocene welded tuff, southern Nevada: Implications for welding temperatures: Journal of Geophysical Research, v. 91, p. 12,817–12,834.

Rosenbaum, J. G., Hudson, M. R., and Scott, R. B., 1991, Paleomagnetic constraints on the geometry and timing of deformation at Yucca Mountain, Nevada: Journal of Geophysical Research, v. 96, p. 1963–1979.

Rosendahl, B. R., Reynolds, D. J., Lorber, P. M., Burgess, C. F., McGill, J., Scott, D., Lambiase, J. J., and Derksen, S. J., 1986, Structural expressions of rifting: Lessons from Lake Tanganyika, Africa, *in* Frostick, L. E., Renaut, R. W., Reid, I., and Tiercelin, J. J., eds., Sedimentation in the African rifts: Geological Society of London Special Publication 25, p. 29–44.

Sawyer, D. A., Fleck, R. J., Lanphere, M. A., Warren, R. G., and Broxton, D. E., 1990, Episodic volcanism in the southwest Nevada volcanic field: New $^{40}Ar/^{39}Ar$ geochronologic results: Eos (American Geophysical Union Transactions), v. 71, p. 1296.

Schlinger, C. M., Rosenbaum, J. G., and Veblen, D. R., 1988, Fe-oxide microcrystals in welded tuff from southern Nevada: Origin of remanence carriers by precipitation in volcanic glass: Geology, v. 16, p. 556–559.

Segall, P., and Pollard, D. D., 1983, Joint formation in granitic rock of the Sierra Nevada: Geological Society of America Bulletin, v. 94, p. 563–575.

Serra, S., and Nelson, R. A., 1988, Clay modeling of rift asymmetry and associated structures: Tectonophysics, v. 153, p. 307–312.

Smith, R. L., 1960, Zones and zonal variations in ash-flow tuffs: U.S. Geological Survey Professional Paper 354-F, p. 149–159.

Stewart, J. H., 1980, Regional tilt patterns of late Cenozoic basin-range fault blocks, western United States: Geological Society of America Bulletin, v. 91, p. 460–464.

——— 1988, Tectonics of the Walker Lane belt, western Great Basin: Mesozoic and Cenozoic deformation in a zone of shear, *in* Ernst, W. G., ed., Metamorphism and crustal evolution of the western United States, Rubey Volume VII: Englewood Cliffs, New Jersey, Prentice-Hall, p. 684–713.

Stewart, J. H., and Poole, F. G., 1974, Lower Paleozoic and uppermost Precambrian Cordillera miogeocline, Great Basin, western United States, *in* Dickinson, W. R., ed., Tectonics and sedimentation: Society of Economic Paleontologists and Mineralogists Special Publication 22, p. 58–82.

Watson, G. S., 1956, Analysis of dispersion on a sphere: Royal Astronomical Society Geophysical Supplement, Monthly Notices, v. 7, p. 153–159.

Wells, R. E., and Hillhouse, J. W., 1989, Paleomagnetism and tectonic rotation of the lower Miocene Peach Springs Tuff: Colorado Plateau, Arizona, to Barstow, California: Geological Society of America Bulletin, v. 101, p. 846–863.

Wernicke, B. P., Axen, G. J., and Snow, J. K., 1988, Basin and Range extensional tectonics at the latitude of Las Vegas, Nevada: Geological Society of America Bulletin, v. 100, p. 1738–1757.

Wright, L., 1976, Late Cenozoic fault patterns and stress fields in the Great Basin and westward displacement of the Sierra Nevada block: Geology, v. 4, p. 489–494.

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